ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 20-Sep-2026

Site: Site Name-1 | Location: Shivamogga, Karnataka | Engine: 100% Old Code Algorithm Match

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
12.18 kW	**20,057 kWh**	**Rs. 571,583**	**4Y 7M**
12.65 kWp Installed	CUF 18.1% | PR 75%	Rs. 5.72 Lakhs	Profit: Rs. 59.17L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Site Name-1
Location / Address	Shivamogga, Karnataka
Contact Email Address	test@voltroof.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	12.18 kW
Total Installed Array Rating	12.65 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	23 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	1440.00 Sq.Feet (133.78 sq.m)
Utilized Usable Rooftop Area	1218 Sq.Feet (113.16 sq.m)
First Year Annual Generation	20,057.08 kWh
25-Year Cumulative Generation	445,625.46 kWh
Estimated Payback Period	4 Years & 7 Months
Post-Payback Profitability (25 Yrs)	Rs. 5,916,514.17 (INR 59.17 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 2,759,432.33 (INR 27.59 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 365 Tonnes CO2 | Tree Equivalence: 584 Mature Trees | Green Energy: 100% Clean LT Power

Based on Central Electricity Authority (CEA) grid emission factor (0.82 kg CO2/kWh) and standard forestry carbon sequestration.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	1440.00 Sq.Feet (133.78 sq.m)
Utilized Effective Rooftop Area	1218 Sq.Feet (113.16 sq.m)
Building Plant Target Capacity	12.18 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	23 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Inverter Rating	13 kW, 415V 3-Ph MPPT

[INVERTER] Inverter Capacity Matrix & Cable Sizing

Grid-tied string inverter selection matrix (Selected rating highlighted)

S.No	Inverter Capacity & Voltage	Qty	Cable Size (Inv-ACDB) / Isolator	Status
01	1 kW, 230V 1-Ph, MPPT	0	-	Alternate
02	1.5 kW, 230V 1-Ph, MPPT	0	-	Alternate
03	2 kW, 230V 1-Ph, MPPT	0	-	Alternate
04	3 kW, 230V 1-Ph, MPPT	0	-	Alternate
05	4 kW, 230V 1-Ph, MPPT	0	-	Alternate
06	5 kW, 230V 1-Ph, MPPT	0	-	Alternate
07	6 kW, 415V 3-Ph, MPPT	0	-	Alternate
08	7 kW, 415V 3-Ph, MPPT	0	-	Alternate
09	8 kW, 415V 3-Ph, MPPT	0	-	Alternate
10	9 kW, 415V 3-Ph, MPPT	0	-	Alternate
11	10 kW, 415V 3-Ph, MPPT	0	-	Alternate
12	11 kW, 415V 3-Ph, MPPT	0	-	Alternate
13	12 kW, 415V 3-Ph, MPPT	0	-	Alternate
14	**13 kW, 415V 3-Ph, MPPT**	**1**	**3.5Cx16 Sq.mm (50A, MCB Type, 4P 415V)**	**SELECTED**
15	14 kW, 415V 3-Ph, MPPT	0	-	Alternate
16	15 kW, 415V 3-Ph, MPPT	0	-	Alternate
17	16 kW, 415V 3-Ph, MPPT	0	-	Alternate
18	17 kW, 415V 3-Ph, MPPT	0	-	Alternate
19	18 kW, 415V 3-Ph, MPPT	0	-	Alternate
20	19 kW, 415V 3-Ph, MPPT	0	-	Alternate
21	20 kW, 415V 3-Ph, MPPT	0	-	Alternate

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear protection, breaker sizing, and cable schedule

Incomer Breaker / Protection	4-Pole MCB / MCCB with thermal magnetic release
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with remote signalling
Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Rated Inverter AC Current	18.09 Amps
ACDB Breaker Current Rating	25 Amps
Cable Size: ACDB to Grid Connection	3.5Cx6 Sq.mm

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550 W mono PERC / TOPCon	23 Nos.	Rs. 230,000
02	On-grid inverter	13 kW, three-phase	1 No.	Rs. 140,833
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 55,000
04	DC solar cable	6 sq.mm copper, PV-rated	130 m	Rs. 11,050
05	AC cable	16 sq.mm copper	70 m	Rs. 16,800
06	MC4 connectors	PV-rated connector pairs	23 pairs	Rs. 6,900
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 12,500
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 12,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 5,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 18,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 55,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 571,583.33**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 5,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 5,000 (Refundable utility escrow)

STATUTORY COMPLIANCE & NET-METERING POLICY

All electrical installations conform to CEA (Technical Standards for Connectivity of Distributed Generation Resources) Regulations 2013 and State Grid Code. Net-metering bi-directional billing operates under applicable KERC / State SERC LT Solar Tariff orders.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 571,583	**4Y 7M**	**Rs. 59.17L**	**Rs. 27.59L**
12-Item BOM Basis	Break-Even Year	Rs. 5,916,514	Alternative Asset

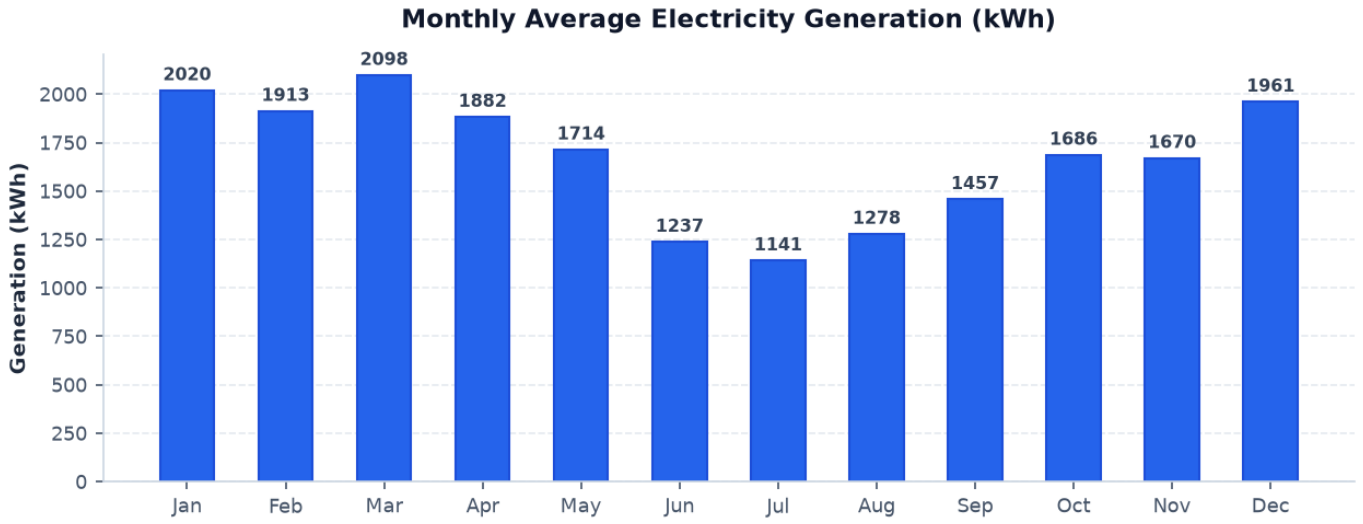
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	20,057	Rs. 8.05	161,459	7,308	154,151	154,151
02	19,857	Rs. 8.45	167,788	7,308	160,480	314,631
03	19,658	Rs. 8.87	174,366	7,308	167,058	481,689
04 *	**19,461**	**Rs. 9.32**	**181,380**	**7,308**	**174,072**	**655,761**
05	19,267	Rs. 9.78	188,429	7,308	181,121	836,882
06	19,074	Rs. 10.27	195,891	7,308	188,583	1,025,465
07	18,883	Rs. 10.78	203,562	7,308	196,254	1,221,719
08	18,695	Rs. 11.32	211,622	7,308	204,314	1,426,033
09	18,508	Rs. 11.89	220,055	7,308	212,747	1,638,780
10	18,322	Rs. 12.48	228,665	7,308	221,357	1,860,136
11	18,139	Rs. 13.11	237,806	7,308	230,498	2,090,634
12	17,958	Rs. 13.76	247,100	7,308	239,792	2,330,426
13	17,778	Rs. 14.45	256,896	7,308	249,588	2,580,015
14	17,601	Rs. 15.18	267,176	7,308	259,868	2,839,883
15	17,425	Rs. 15.93	277,572	7,308	270,264	3,110,147
16	17,250	Rs. 16.74	288,769	7,308	281,461	3,391,608
17	17,078	Rs. 17.57	300,056	7,308	292,748	3,684,356
18	16,907	Rs. 18.45	311,934	7,308	304,626	3,988,982
19	16,738	Rs. 19.37	324,213	7,308	316,905	4,305,887
20	16,571	Rs. 20.34	337,045	7,308	329,737	4,635,624
21	16,405	Rs. 21.36	350,407	7,308	343,099	4,978,723
22	16,241	Rs. 22.43	364,281	7,308	356,973	5,335,696
23	16,078	Rs. 23.55	378,646	7,308	371,338	5,707,033
24	15,918	Rs. 24.73	393,642	7,308	386,334	6,093,367
25	15,758	Rs. 25.96	409,088	7,308	401,780	6,495,148

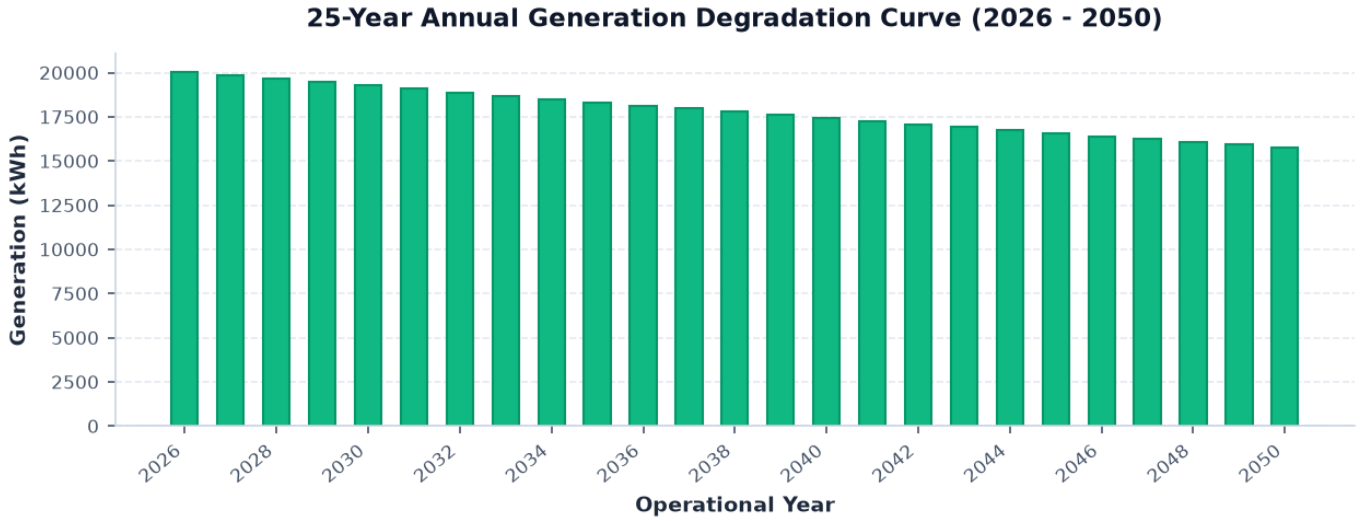
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1647 kWh/kWp	75.0%	18.1%	1.0% / year
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

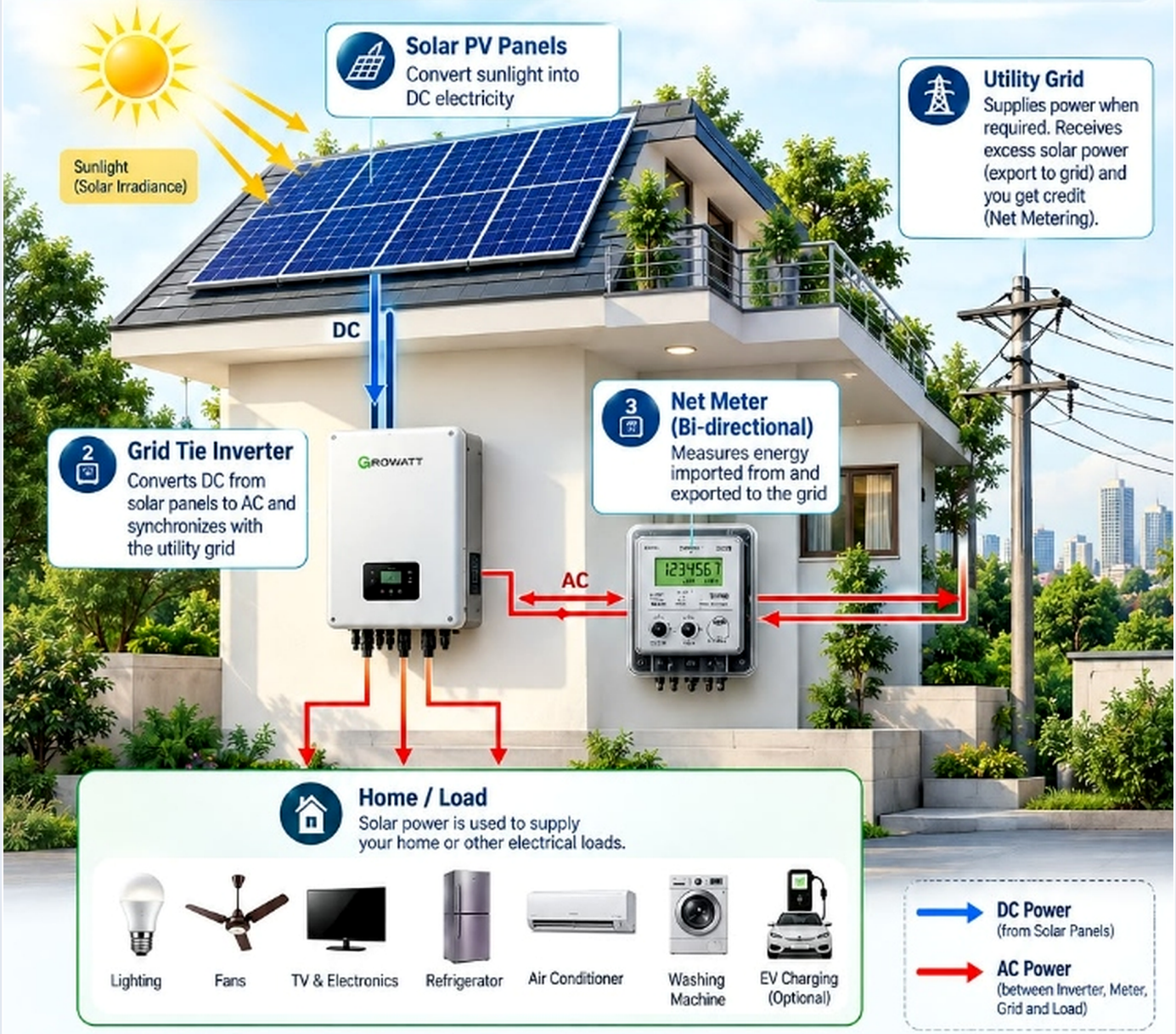
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow